LCC RPiPico Node (v3.0)

An open, DIY-friendly Raspberry Pi Pico platform for OpenLCB / LCC model railroad control

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The LCC RPiPico Node is a single, general-purpose board design that replaces a whole family of earlier dedicated boards. Specialized functions — displays, stepper drivers, servos, and more — plug in as breakout boards rather than requiring a new board revision for every node type. It grew out of LocoNet turntable controllers dating back to 2021 and migrated to full LCC/OpenLCB support in 2024.

WHY IT'S AN EASY DIY ENTRY POINT

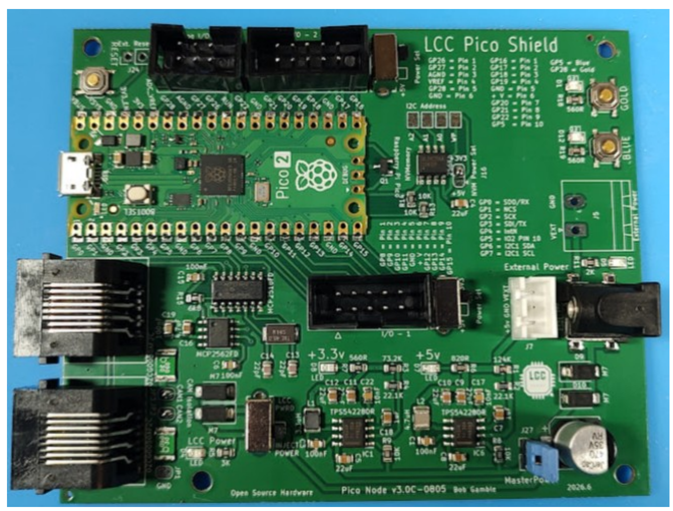
- **Arduino IDE** — programmed with the standard Earle Philhower RP2040 board package.
- **Full LCC/OpenLCB** stack included via the OpenLcbCLib library — alias allocation, datagrams, CDI, memory config, and more already implemented.
- **One board**, every node type — swap breakout boards instead of re-designing hardware for each new function.
- **Hand-solderable** — discrete MCP2518/MCP2562 CAN chips instead of hard-to-hand-solder integrated parts.
- **Documented pin/architecture standard** shared across every example project, so patterns learned on one node transfer directly to the next.
- **Working example firmware to copy** — four complete, real-world node applications with full source (see below).

EXAMPLE NODE PROJECTS *(OPEN SOURCE, FULL FIRMWARE ON GITHUB)*

- Turntable — Motorized turntable controller — stepper-driven positioning, home/limit sensing, touchscreen UI with track selection, bridge animation, and RailCom occupancy display.
- Roundhouse — Multi-stall roundhouse door control — servo-driven doors synchronized to turntable track selection.
- PixelLights — Addressable NeoPixel lighting controller — scene/effect control for structure, signal, or ambient layout lighting over LCC events.
- Clock Lights — Layout lighting tied to the LCC fast clock — day/night lighting transitions and dimming driven by the shared broadcast fast-clock protocol.

KEY BOARD ATTRIBUTES

- **Processor:** Raspberry Pi Pico or Pico 2 (Pico 2 recommended)
- **CAN:** discrete MCP2518 controller + MCP2562 transceiver
- **Power:** dual 3.3V/5V switching regulation; power from LCC bus or 8–15V external; can inject power onto the bus
- **NVM:** onboard I2C EEPROM or FRAM spares the Pico's flash from wear
- **I/O:** 16 GP pins via two 10-pin IDC connectors (I/O-1, I/O-2) + a 6-pin analog header (I/O-3)
- **Breakout ecosystem:** touchscreen displays (SPI or parallel), stepper driver, servo, and test/ribbon breakouts
- **Buttons:** onboard reset (Pico has none) + header for a remote reset button; Blue/Gold (shared pins)



GET STARTED

- **Board hardware** (documentation, schematics, PCB, gerbers, breakout designs): **github.com/bocabob/LCC-RPi-Pico-Board**
- **Firmware architecture standard & example application links: github.com/bocabob/LCC_RPiPico_Common**
- **Board support:** Earle Philhower's RP2040 core (arduino-pico) —
  install instructions at **arduino-pico.readthedocs.io/en/latest/install.html**
- **LCC protocol stack:** OpenLcbCLib at **github.com/JimKueneman/OpenLcbCLib**
  or OpenLCB Single Thread at **github.com/openlcb/OpenLCB_Single_Thread**